Sensing-Assisted Turbulence-Induced Secrete Key Distribution

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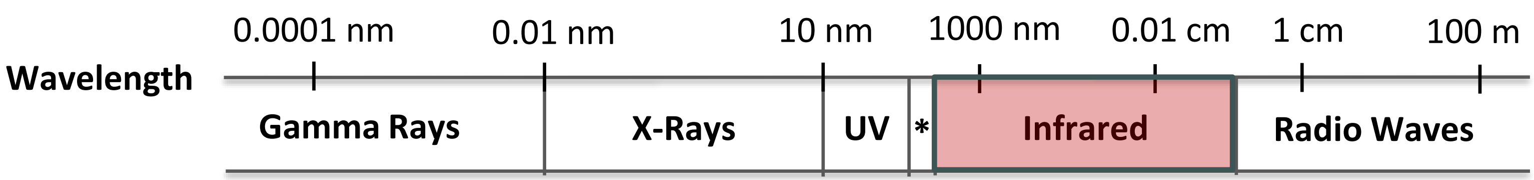
Outline of presentation

Free-Space Optics (FSO): Advantages and Challenges

○ For high-speed connections, FSO is a promising candidate

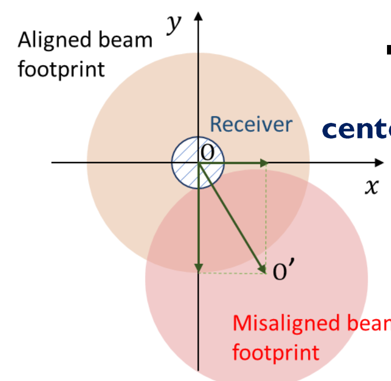
- Using infrared bands → support extremely high data rates (~ Gbps, up to Tbps)
- Low power consumption and License-free spectrum

(*): Visible light



○ Challenges in implementation of FSO

- Atmospheric turbulence, Atmospheric attenuation
- Pointing errors: Cause a decrease and fluctuation in the received signal
  → Degrade the performance of FSO systems



- **Misalignment between beam center and detector**

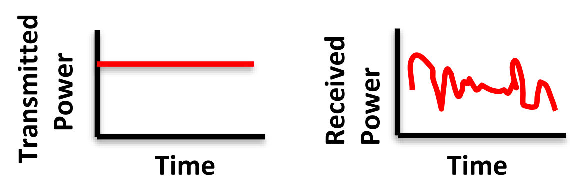
\- Pointing errors are caused by:

\+ Mechanical vibrations of transceiver platforms (e.g., UAVs, satellites)

\+ Imperfect beam tracking and alignment systems

\+ Installation errors causing boresight offset

\+ Atmospheric turbulence causing beam wander



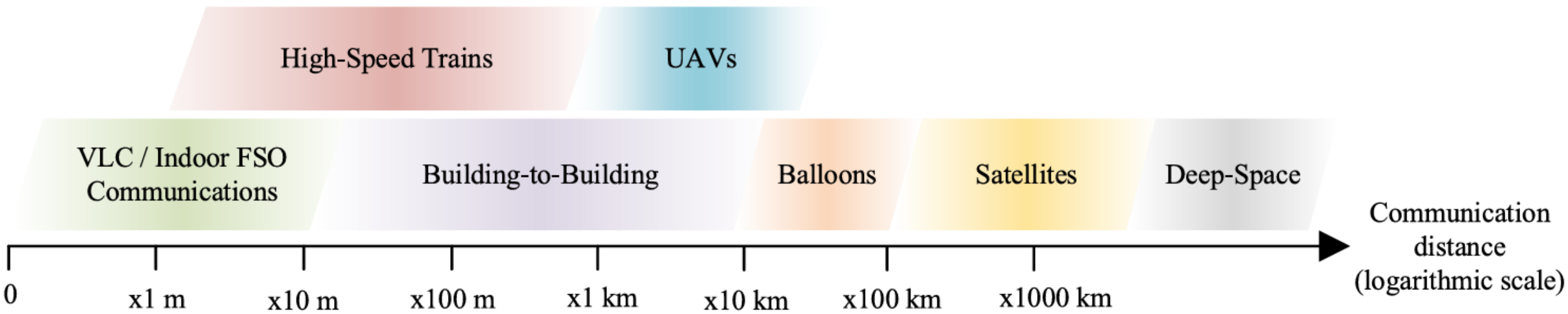
What is the solutions for Pointing errors?

Pointing errors solution – overview of APT mechanisms

- **Definition of APT mechanisms:**
  - An Acquisition, Pointing, and Tracking (APT) mechanism is a system used in FSO communications to establish, align, and maintain a reliable line-of-sight optical link between transceivers
  - APT mechanisms can be classified according to working principles and use cases [1]

+ Working principles: Gimbal-based, Mirror-based, Gimbal-Mirror Hybrid, Adaptive Optics, Liquid Crystal, RF-FSO Hybrid, and other alternative ATP mechanisms.

+ Used cases: Building-to-Building, High-Speed Trains, UAVs, Balloons, Satellites, and Deep Space.

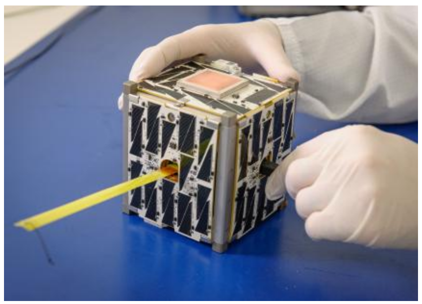


[1]: Kaymak, Yagiz, et al. "A survey on acquisition, tracking, and pointing mechanisms for mobile free-space optical communications." *IEEE communications surveys & tutorials* 20.2 (2018): 1104-1123.

Limitation of APT mechanisms and Alternative solution

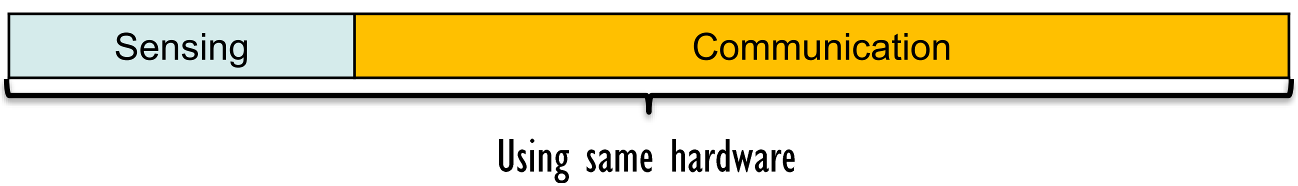
- **Limitation:**
  - Traditional APT systems often suffer from high complexity, power consumption, and payload requirements [2]

  → Making them less suitable for lightweight UAVs and small satellite (CubeSats)



→ A CubeSat built at NASA's Ames Research Center measures about 10 centimeters in a side.

- **Alternative solution: Sensing-assisted optical wireless communication**
  - Key idea: Time-division multiplexing is used to allow sensing and the communication to share the same transceiver hardware, where sensing phase acts as the function of an APT system.



→ Reduce resource consumption → Suitable for lightweight UAVs and small satellites

[2]: Li, Yuechao, et al. "Sensing assisted optical wireless communication for UAVs." IEEE Transactions on Vehicular Technology 73.12 (2024): 18620-18634.

Example: Corner reflector and Modulating Retroreflector (MRR)

- Corner reflector and MRR are widely use in "sensing" in FSO communication

1. Corner reflector

- Definition: Corner reflector is a device that reflects a signal or light back to where it came from, and it is usually made of three perpendicular reflecting surfaces.

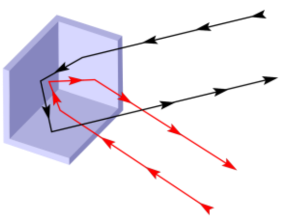
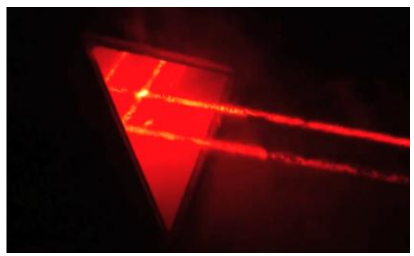
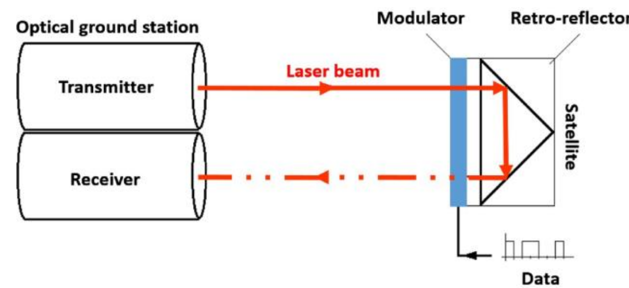



Fig. 1. Illustration of a corner reflector

2. MRR: Definition

- MRR is a passive optical device (Including corner reflector and a modulator) that reflects an incoming laser beam back toward its source.
- It modulates the reflected signal by using a high-speed optical shutter placed in front of the reflector to encode information.

→ MRR can help address the power constraints of small satellites.



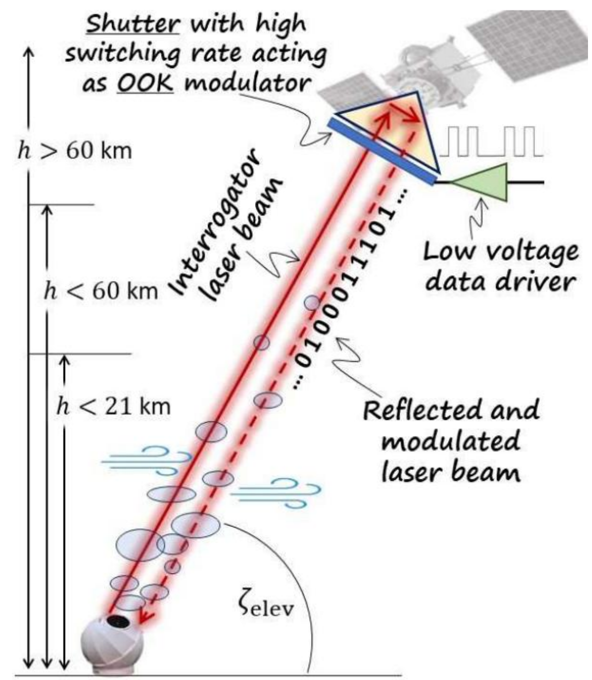


Fig1. Illustration of MRR

Outline of presentation

I. Introduction

II. System model

III. Conclusion

Sensing assisted UAV-to-Ground Station communication

➢ System model:

- Bob: A hovering UAV without an APT system, using sensing to determine the location of BS (Alice)
- Alice: Base station (BS) equipped with an APT system
- Alice and Bob extract the randomness of the channel to generate secret keys
- Eve: Passive eavesdropper located within the beam footprint of Bob

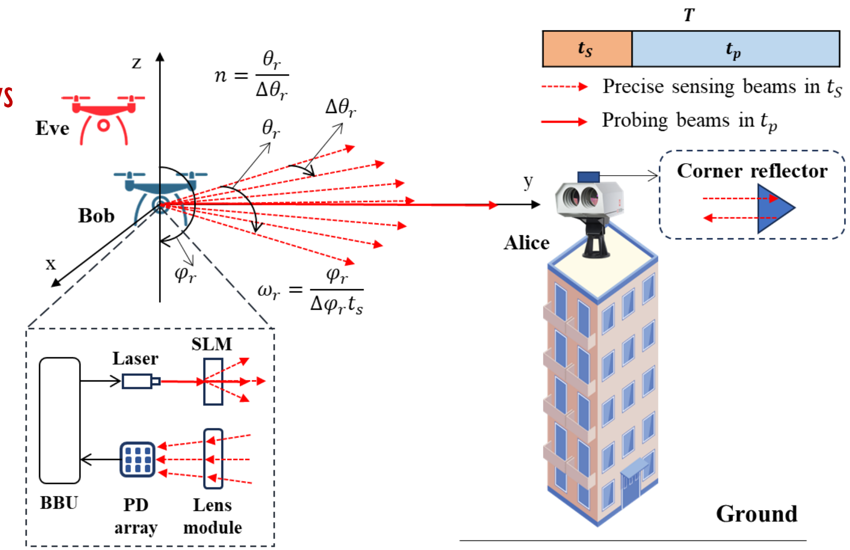


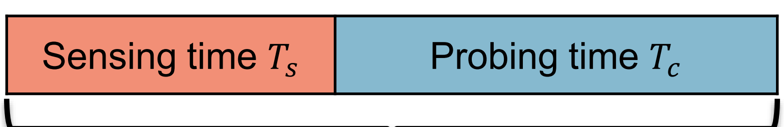
Fig2. System model

➢ Channel model:

- Forward channel h_{AB} : Atmospheric turbulence, pointing error (jitter)
- Reverse channel h_{BA} : Atmospheric turbulence, pointing error (jitter + sensing error)

➢ Proposed scheme:

→ Sensing-Assisted Turbulence-Induced Secrete Key Distribution scheme: Time-division multiplexing is used to allow sensing and the probing time to share the same transceiver hardware



$$T_{Communication} = T_s + T_c$$

- Sensing phase: UAV (Bob) accurately estimates the receiver's position (Alice)
- Probing phase: Alice and Bob extract the randomness of the turbulence channel to generate secret keys

Sensing assisted UAV-to-ground station

➢ **Key process:**

Raw key generation

- Step I. Channel sensing: UAV (Bob) accurately estimates the receiver's position (Alice)

- Step II. Channel probing: Alice and Bob exchange pilot signals to observe the random channel properties

- Step III. Quantization: The measured values (channel gain) are converted into binary bits (0 and 1)

Post-processing

- Step IV. Key Reconciliation: Errors are corrected (by ECC) so both keys are identical.

- Step V. Privacy Amplification: The key is refined to remove any information an eavesdropper might have.

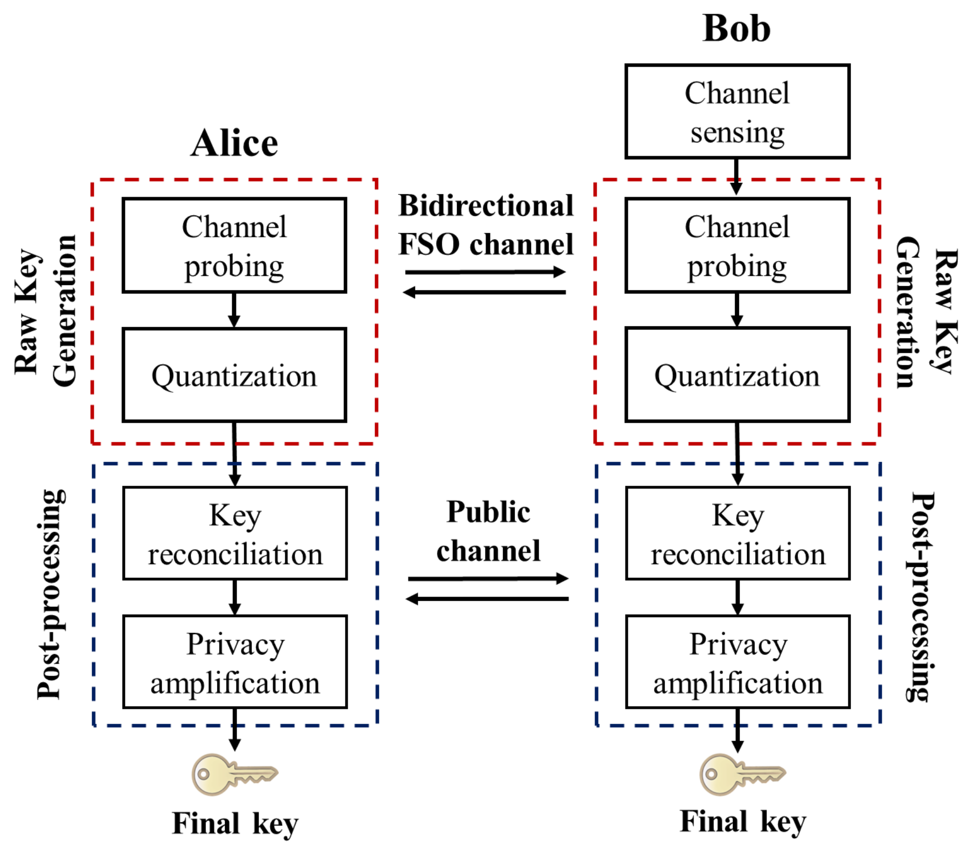


Fig3. Key process

Sensing assisted UAV-to-ground station

Sensing phase:

- Target: Bob Estimate azimuth and pitch angles of BS (Alice)
- Method: multi-line (Horizonal scanning) and rotational optical scanning (vertical scanning)

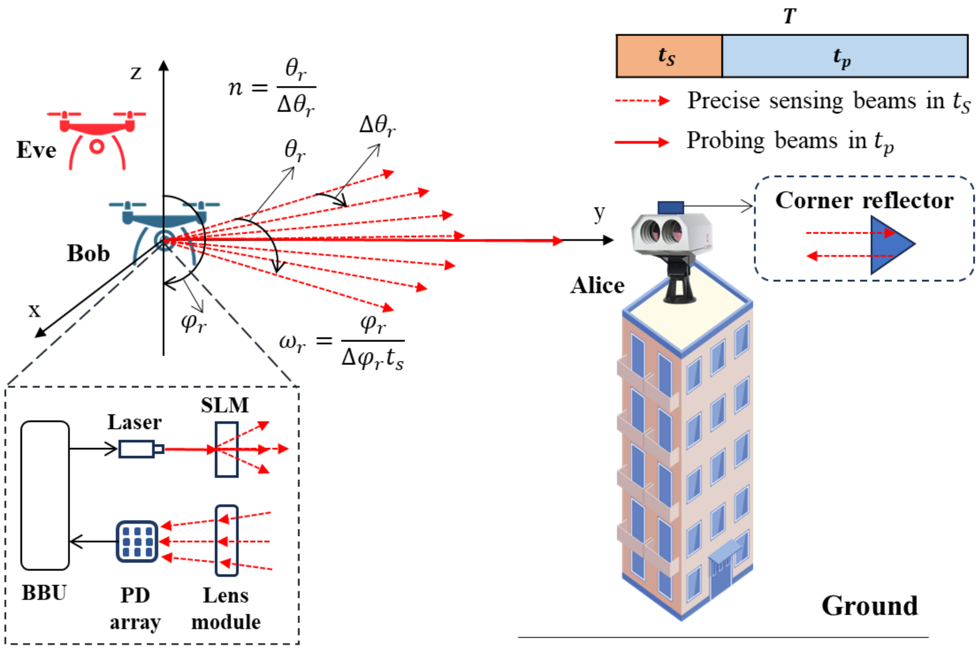
❑ Horizontal scanning → azimuth angle

- Horizontal scanning range: θ_r
- Number of lines in the scanning: n
- Horizontal scanning step: $|\triangle\ \theta_r| = \theta_r/n$

❑ Vertical scanning → pitch angles

- Vertical scanning range: φ_r
- Vertical scanning step: $\triangle\ \varphi_r = \varphi_r/(\omega_r t_S)$

Where: the scanning frequency (ω_r) and sensing time (t_S)



→ The UAV estimates the location of the BS using reflected signals from the corner reflector collected by the PD array.

Sensing assisted UAV-to-ground station

- **Sensing phase:**

I. When UAV without jitter (only sensing error)

- The estimated position of the lens center can be expressed as:

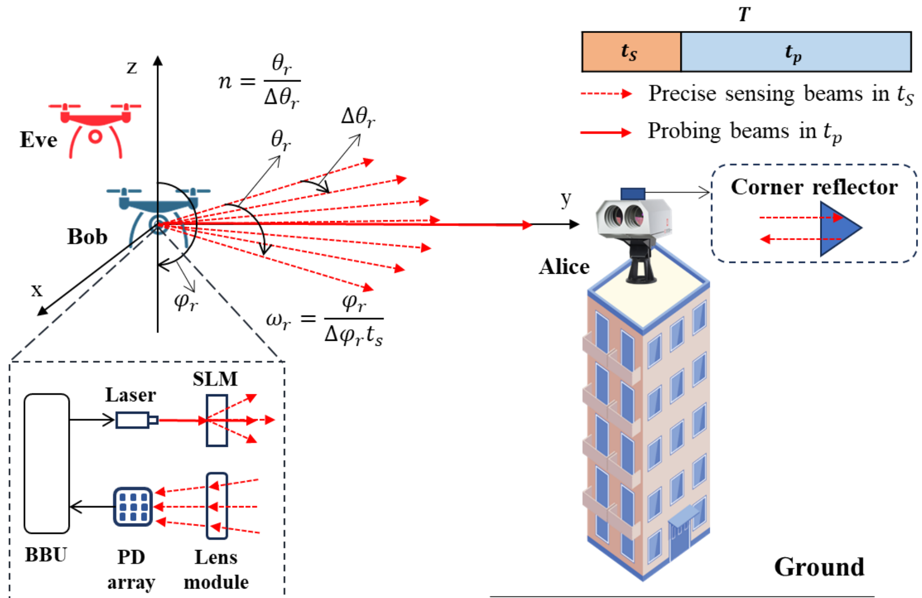
$$x = Dtan(\triangle\, \theta_r) = D \triangle \theta_r$$

$$z = D\frac{tan\triangle\varphi_r}{cos\triangle\theta_r} = D \triangle \varphi_r$$

$$y = D$$

- $\triangle\, \theta_r$: the horizontal scanning step
- $\triangle\, \varphi_r$: the vertical scanning step (depend on sensing time)

→ The radial distance between the spot center and the lens center can be obtained as: $r = \sqrt{x^2 + z^2}$



II. When UAV with jitter (sensing error+ Jitter) [2]

$$x' \approx D \triangle \theta_r + \triangle\, x$$
$$z' \approx D \triangle \theta_r + \triangle\, z$$

Where: $\triangle\, x$; $\triangle\, z \sim \mathcal{N}(0, \sigma^2)$: The displacement deviation caused by jitter

→ The radial distance between the spot center and the lens center can be obtained as: $r = \sqrt{x'^2 + z'^2}$

[2]: Li, Yuechao, et al. "Sensing assisted optical wireless communication for UAVs." IEEE Transactions on Vehicular Technology 73.12 (2024): 18620-18634.

Simulation results

Step I: Data generation under the impact of sensing time

Step II: Secret key rate (SKR) evaluation

→ 1. Evaluate the SKR under the impact of different channel conditions in the Alice–Bob link (different levels of turbulence and levels of channel correlation of turbulence between the forward and reverse links), as well as sensing time.

→ 2. Evaluate the SKR under the impact of different eavesdropper conditions, including the turbulence correlation between Eve's channel and the Alice–Bob channel, and the eavesdropper's location.

Simulation results- Step 1: Data generation

1. Time-Correlated Turbulence Channel Generation [3] - $h_{turbulence}(t)$

- The channel correlation between turbulence channel of forward and inverse link is ρ

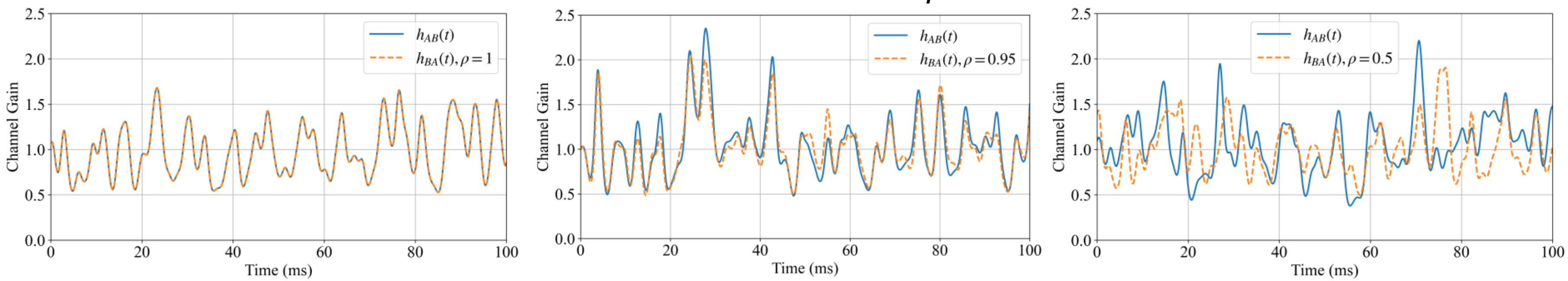


Fig1. Time series of correlated turbulence channel gains for forward and reverse links with ρ

2. Time-Correlated Pointing Error Channel Model [4]- $h_p(t)$

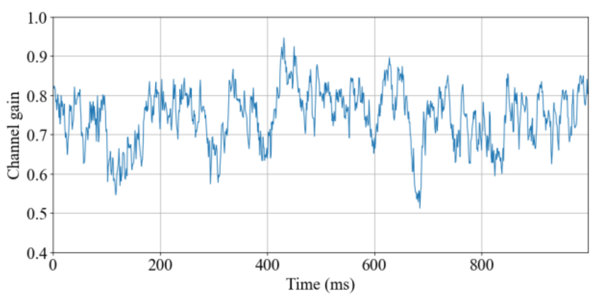


Fig2. Time series of correlated pointing error channel gain

3. Total channel [4]- $h(t) = h_p(t).h_{turbulence}(t)$

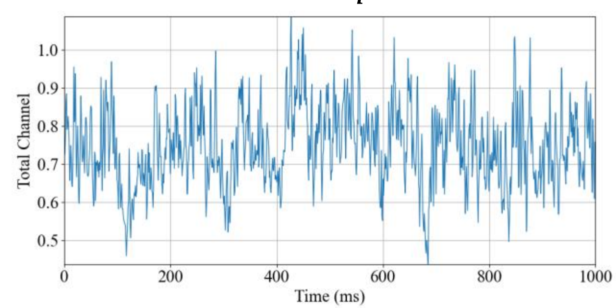


Fig3. Time series of the total correlated channel gain.

[3]: S. Galijasevic, J. Luo, D. Divsalar, and R. Wesel, "Channel-prediction-driven rate control for ldpc coding in a fading fso channel with delayedfeedback," IEEE Open Journal of the Communications Society, vol. 6,pp. 3320–3331, 2025.

[4]: . W. Sun and J. H. Noh, "End-to-end performance analysis of ccsds o3koptical communication system under atmospheric turbulence and pointingerrors," Aerospace, vol. 12, no. 10, p. 869, 2025.

Simulation results

➢ Step II: SKR estimation

1. Quantization step:

- Purpose: Converts analog channel measurements into binary values (preliminary key)
- Quantization Types:

+ lossy quantization: Discards ambiguous samples → Increase Key Disagreement Rate - KDR; Key Generation Rate - KGR)

+ lossless quantization: Keeps all samples → Reduce KDR, Maximize KGR

2. Scheme: Two-Threshold Lossy Quantization [5]

The estimate channel at receiver is h

$$q^+ = mean(h) + \alpha.\sigma(h)$$

$$q^- = mean(h) - \alpha.\sigma(h)$$

Where: $mean(h)$: average of the channel estimates;

$\sigma(h)$: standard deviation; and α: scaling factor.

$$\rightarrow \begin{cases} h_i > q^+ \rightarrow h_i = 1 \\ h_i < q^- \rightarrow h_i = 0 \\ q^- < h_i < q^+ \rightarrow discard \end{cases}$$

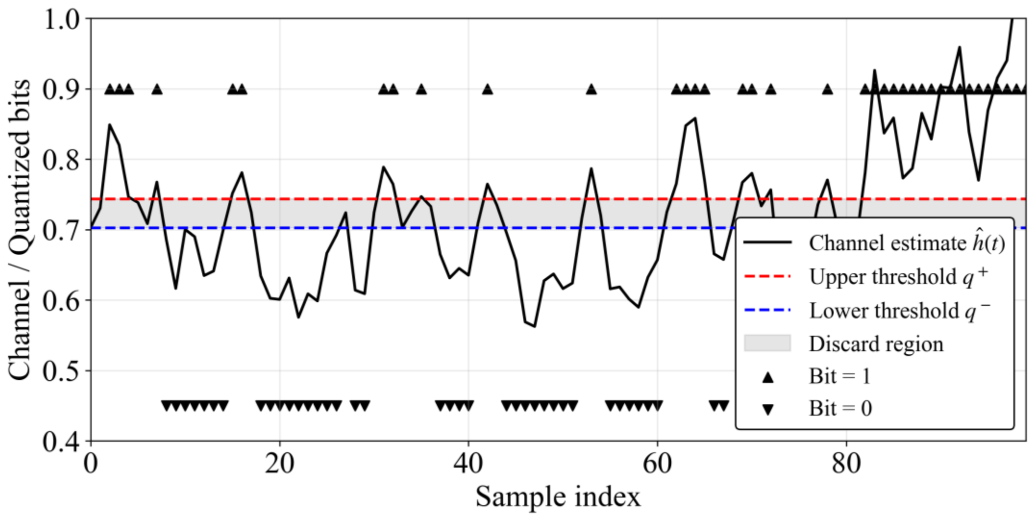


Fig4. Bit extraction using the two-threshold quantization scheme.

[5]: Alshamdayn, Abdullah Dakhlallah, and Manki Min. "A Survey of Quantization Techniques for Physical Layer-Based Secret Key Generation in IoT Systems." 2024 IEEE International Conference on Computing (ICOCO). IEEE, 2024.

Simulation results

- SKR estimation - without passive eavesdropper

- SKR under impact of different channel condition and sensing time

* Strong week = Very week

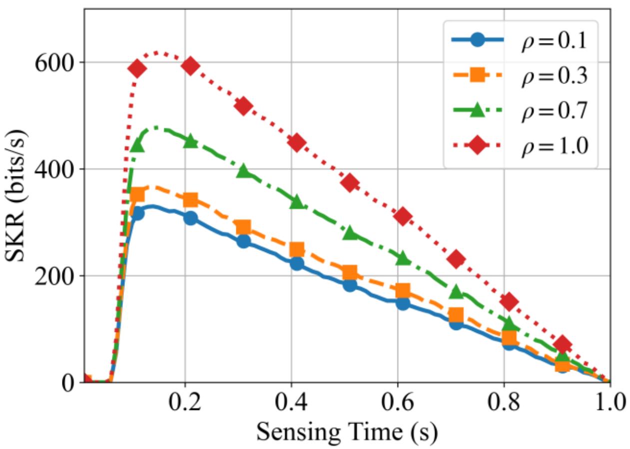


Fig. 5. SKR vs. Sensing Time under varying channel correlations ρ (Weak turbulence).

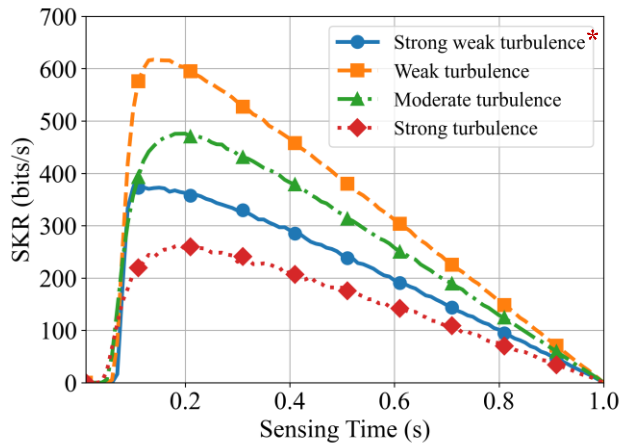


Fig. 6. SKR vs. Sensing Time across different turbulence levels ($\rho = 0.9$).

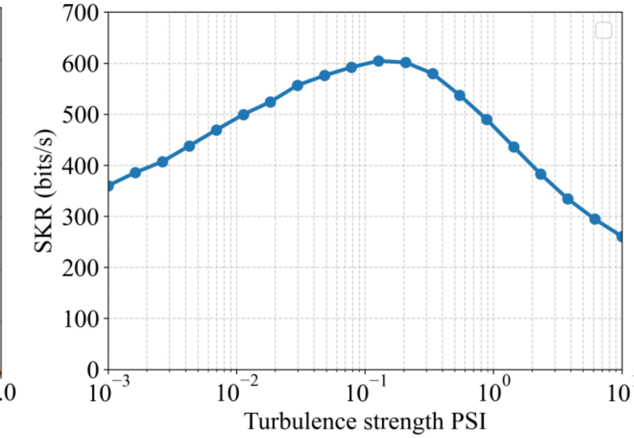


Fig. 7. SKR vs. Turbulence Strength (PSI) at an optimal sensing time $t_s = 0.2$

→ Key Insights:

- Optimal Sensing Time: There is a clear trade-off; an optimal sensing time (around 0.15s - 0.2s) exists to maximize the SKR across all conditions.
- Correlation Impact (Fig. 1): Higher channel correlation between ρ Alice and Bob significantly improves the maximum achievable SKR.
- Turbulence Impact (Fig. 2 & 3): SKR drops in extreme conditions: "too strong" destroys the signal, while "too weak" lacks randomness for key generation. Weak to moderate turbulence is ideal for achieving the highest SKR

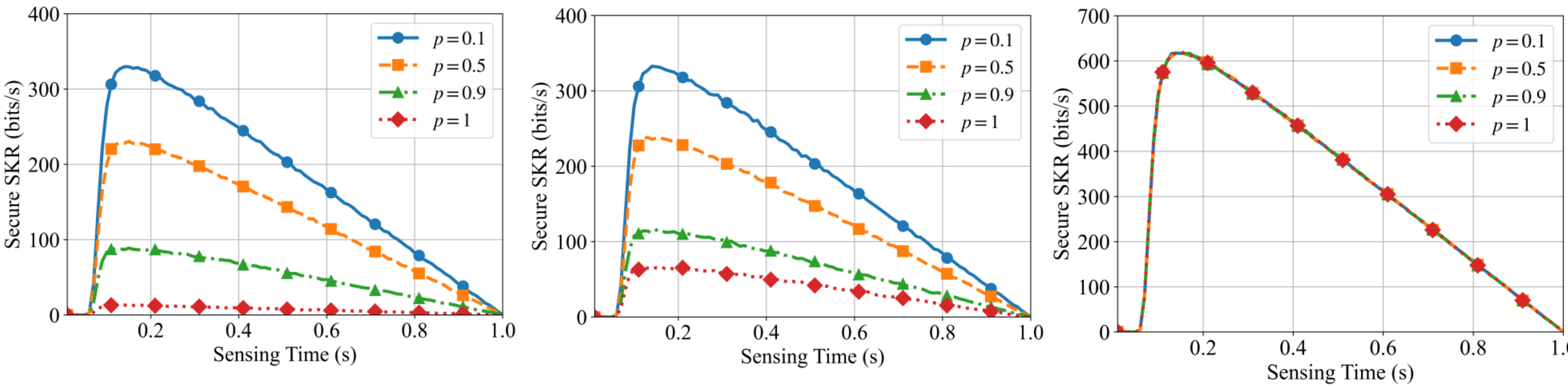
Level of turbulence	PSI (Power Scintillation Index [3])
Strong weak*	$PSI = 10^{-3}$
Weak	$PSI = 10^{-1}$
Moderate	$PSI = 10^{0}$
Strong	$PSI = 10^{1}$

[3]: S. Galijasevic, J. Luo, D. Divsalar, and R. Wesel, "Channel-prediction-driven rate control for ldpc coding in a fading fso channel with delayedfeedback," IEEE Open Journal of the Communications Society, vol. 6,pp. 3320–3331, 2025.

Simulation results

➢ **Step II: SKR Estimation - Presence of a passive eavesdropper (Eve)**

- Context: Assuming perfect Alice-Bob channel correlation under weak turbulence.



a. $d_{BE} = 0.2\ beamwidth$ b. $d_{BE} = 0.5\ beamwidth$ c. $d_{BE} = 1.5\ beamwidth$

Fig. 8. Secure SKR vs. Sensing Time for varying Eve channel correlations (ρ) at different Bob-Eve distances (d_{BE}).

→ Key Insights:

- Impact of Eve's Location d_{BE} : Secure SKR increases when Eve is positioned further from the beam center
- Impact of Channel Correlation (ρ): Secure SKR significantly increases when Eve's channel is less correlated with the legitimate channel

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Conclusion

- Proposed a sensing-assisted Secret Key Distribution (SKD) scheme for UAV-BS communications.
- Developed a comprehensive simulation model to evaluate the Secret Key Rate (SKR) under various channel conditions.
- Investigated the impact of sensing time and solved the optimal time allocation problem to maximize system performance.

Thank you for your listening!